MIL-A-11186A (MU)
26 January 1968
SUPERSEDING
MIL-A-11186 (ORD)
31 May 1951

MILITARY SPECIFICATION

ADAPTER, GRENADE PROJECTION,
CHEMICAL, M2A1

1. SCOPE

1.1 This specification covers the metal parts assembly for one type of adapter designated as Adapter, Grenade Projection, Chemical, M2A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions

FSC: 1330

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MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9232012 - Adapter, Grenade Projection, Chemical, M2A1 Assembly
9232107 - Carton, Packing, Ammunition, for Adapters, Grenade Projection, Chemical, M2A1
9232108 - Box, Packing, Ammunition, for Adapters, Grenade Projection, Chemical, M2A1

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 9232012 - Equipment Tabulation Number

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Metal parts assembly.- The metal part assembly shall comply with all requirements specified on drawing (dwg.) 9232012 and with all requirements specified in applicable specifications.

3.3 Strength of stabilizer tube brazed joint.- The brazed joint of the stabilizer tube and plug assembly shall not crack or show evidence of separation when tested as specified in 4.4.1.

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3.4 Strength of stabilizer tube and fin assembly welds.- The spot welds shall leave a button of fin material on the stabilizer tube, which is the thickness of the fin material and has an area at the base of at least one-half the area of the specified weld diameter, when tested as specified in 4.4.2.

3.5 Embrittlement of springs.- Setback band springs shall not break when extended to the minimum (min.) point of permanent distortion when tested as specified in 4.4.3.

3.6 Functioning

3.6.1 Action at launching.- The setback band shall release the fuze safety lever when the adapter is fired from a standard rifle and grenade launcher as specified in 4.4.6.

3.6.2 Action in flight.- The grenade and adapter, shall remain attached throughout their free flight, when fired as specified in 4.4.6.

3.7 Workmanship.- All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

3.8 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such

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inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define the terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.- Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

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4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 assemblies and 25 complete sets of parts (i.e. 25 each of every component and every sub-assembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. The 25 assemblies shall be tested in a Government Proving Ground. All parts and materials, including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials, (see 4.1.1e). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.- Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any component or assembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the

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Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain parts of one kind from one supplier. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components, or sub-assemblies other than items for delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examinations.- Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 except that inspection for critical defects, (and majors, when so specified) shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 for defects other than critical may be used if approved by the procuring activity. Also, at the option of the procuring activity AQL's and sampling plans may be applied to

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the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Tube, stabilizer (see dwg. 9216144 covering a detail of dwg. 9232011).

Categories	Defects	Method of Inspection	Code No. (see 6.2
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Wall thickness, minimum (min.)..	Gage	01001
Minor:	AQL 1.50 percent		
201.	Diameter of flared flange, min..	Gage	01002
202.	Length of tube, min.....	Gage	01003
203.	Evidence of poor workmanship (see 3.7).....	Visual	01004

4.3.2.2 Fin (see dwg. 9216137 covering a detail of dwg. 9232012).

Categories	Defects	Method of Inspection	Code No
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Crack or fracture.....	Visual	02001
102.	Spot weld missing or of incorrect size (when applicable).....	Visual	02002
Minor:	AQL 1.50 percent		
201.	Inside diameter, min.....	Gage	02003
202.	Width of ring either side of formed section.....	Gage	02004
203.	Total length.....	Gage	02005
204.	Evidence of poor workmanship (see 3.7).....	Visual	02006

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4.3.2.3 Clip (see dwg. 9232008 covering a detail of dwg. 9232012).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Angle of prong	Gage	03001
102.	Width of prong	Gage	03002
103.	Length from center to inside of prong	Gage	03003
104.	Length to large radius	Gage	03004
Minor:	AQL 1.50 percent		
201.	Spacing of prongs	Gage	03005
202.	Total length after forming	Gage	03006
203.	Inside diameter	Gage	03007
204.	Evidence of poor workmanship (see 3.7)	Visual	03008

4.3.2.4 Plug (see dwg. 9232009 covering a detail of dwg. 9232011).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Outside diameter at open end	Gage	04001
102.	Length to shoulder from open end	Gage	04002
103.	Total length	Gage	04003
Minor:	AQL 1.50 percent		
201.	Length to shoulder from closed end	Gage	04004
202.	Brazing ring diameter	Gage	04005
203.	Length of brazing ring diameter.	Gage	04006
204.	Evidence of poor workmanship (see 3.7)	Visual	04007

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4.3.2.5 Base (see dwg. 9232007 covering a detail of dwg. 9232012).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Diameter of hole	Gage	05001
202.	Thickness	Gage	05002
203.	Outer diameter	Gage	05003
204.	Evidence of poor workmanship (see 3.7)	Visual	05004

4.3.2.6 Washer (see dwg. 9232006 covering a detail of dwg. 9232012).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Diameter of hole	Gage	06001
202.	Thickness	Gage	06002
203.	Outer diameter	Gage	06003
204.	Evidence of poor workmanship (see 3.7)	Visual	06004

4.3.2.7 Clip, launcher positioning (see dwg. 9216145).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Clip failure, functional	Gage	07001

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Minor:	AQL 1.50 percent		
201.	Length after forming (max.).....	Gage	07002
202.	Width.....	Gage	07003
203.	Evidence of poor workmanship (see 3.7).....	Visual	07004

4.3.2.8 Stabilizer tube and plug assembly (see dwg. 9232011 covering a detail of dwg. 9232012).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Total length, max.....	Gage	08001
Minor: None defined			

4.3.2.9 Spring, setback (see dwg. 9232004 covering a detail of dwg. 9232010).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Spring loop gap, max.....	Gage	09001
Major:	AQL 1.00 percent		
101.	Outside diameter, max.....	Gage	09002
102.	Load at extended length.....	Test	09003
103.	Rate per inch.....	Test	09004
104.	Initial tension.....	Test	09005
105.	Free length.....	Gage	09006
106.	Protective coating of spring damaged.....	Visual	09007
Minor:	AQL 1.50 percent		
201.	Formation of loop improper.....	Visual	09008
202.	Position of loop inaccurate.....	Gage	09009
203.	Evidence of poor workmanship (see 3.7).....	Visual	09010

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4.3.2.10 Band, setback, prior to application of protective coating (see dwg. 9232005 covering a detail of dwg. 9232010).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack or split.....	Visual	10001
Major:	AQL 0.40 percent		
101.	Distance between holes.....	Gage	10002
102.	Overall length, max.....	Gage	10003
Minor:	AQL 1.50 percent		
201.	Overall width.....	Gage	10004
202.	Thickness.....	Gage	10005
203.	Evidence of poor workmanship (see 3.7).....	Visual	10006

4.3.2.11 Setback band assembly (see dwg. 9232010 covering a detail of dwg. 9232012).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Loops of spring not properly assembled in holes of setback band.....	Visual-Manual	11001
Major:	None defined		
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7).....	Visual	11002

4.3.2.12 Assembly, prior to application of protective coating (see dwg. 9232012).

Categories	Defects	Method of Inspection	Code No.

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Critical:			
1.	Base crimp inadequate.....	Visual-Manual	12001
2.	Clip split or cracked.....	Visual	12002
Major:	AQL 1.00 percent		
101.	Stabilizer tube split or cracked.....	Visual	12003
102.	Inside diameter of stabilizer tube.....	Gage	12004
103.	Concentricity of fin to tube....	Gage	12005
104.	Fin insecure.....	Manual	12006
105.	Straightness, full length.....	Gage	12007
Minor:	AQL 1.00 percent		
201.	Fin weld missing.....	Visual	12008
202.	Total length.....	Gage	12009

4.3.2.13 Assembly (see dwg. 9232012).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Exterior protective coating (except spring) inadequate.....	Visual	13001
202.	Marking incomplete, misleading or unidentifiable.....	Visual	13002
203.	Preservative oil missing from interior of stabilizer tube.....	Visual	13003
204.	Evidence of poor workmanship (see 3.7).....	Visual	13004

4.3.2.14 Carton, prior to sealing (see dwg. 9232107).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Categories	Defects	Method of Inspection	Code No.
Major:	AQL 0.65 percent		
101.	Incorrect number of adapters....	Visual	14001
102.	Setback band missing.....	Visual	14002
103.	Positioning clip missing.....	Visual	14003
104.	Cartridges missing.....	Visual	14004
Minor:	None defined		

4.3.2.15 Carton, sealed (see dwg. 9232107).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Sealing strip incomplete or badly wrinkled.....	Visual	15001
102.	Assembly damaged so that contents are exposed or liable to become exposed.....	Visual	15002
103.	Sealing compound incomplete.....	Visual	15003
Minor:	AQL 1.00 percent		
201.	Contents move when carton is shaken.....	Manual	15004
202.	Label missing, misleading or marking unidentifiable.....	Visual	15005

4.3.2.16 Packing box (see dwg. 9232108).

Categories	Defects	Method of Inspection	Code No
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Contents of box incomplete.....	Visual	16001
102.	Box damaged.....	Visual	16002
103.	Hardware missing, broken or loose.....	Visual-Manual	16003
104.	Strapping missing, broken, loose.	Visual-Manual	16004

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Minor:	AQL 1.50 percent		
201.	Hardware improperly engaged.....	Visual	16005
202.	Strapping improperly assembled..	Visual	16006
203.	Seal missing, unsealed, or improperly positioned.....	Visual	16007
204.	Marking incorrect, incomplete or unidentifiable.....	Visual	16008

4.3.3 Testing

4.3.3.1 Stabilizer tube, brazed joint

4.3.3.1.1 1000 pound load (see 3.3) - Major defect - Code No. 17001.- This test shall be performed one hundred percent. Any brazed joint which shows evidence of failure shall cause the assembly to be classed defective. The test shall be performed as specified in 4.4.1.1 using equipment in accordance with 4.3.4.

4.3.3.1.2 7500 pound load test (see 3.3) - Major defect - Code No. 18001.- The sampling plan for this test shall be in accordance with MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. Any stabilizer tube assembly which fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.1.2 using equipment in accordance with 4.3.4. Samples used in this test shall not be returned to the lot.

4.3.3.2 Strength of stabilizer tube and fin assembly welds (see 3.4) - Major defect - Code No. 19001. Samples used in this test shall not be returned to the lot.

4.3.3.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have been found to be acceptable, eighty (80) assemblies shall be selected from each lot for test. If two (2) or more assemblies show evidence of failure, the lot shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.2.2 After three (3) consecutive lots have complied with the criteria of 4.3.3.2.1, the sampling plan for this test shall be in accordance with MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. Any stabilizer tube and fin assembly spot weld which

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fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.3 Embrittlement of springs (see 3.5) - Major defect - Code No. 20001.- Thirty (30) springs shall be selected from each lot for this test. The test shall be performed as specified in 4.4.3. If one or more springs fail to comply with the specified requirements, the lot shall be rejected. Samples used in this test shall not be returned to the lot.

4.3.3.4 Hardness of clip (see dwg. 9232008) - Major defect - Code No. 21001.- Twenty (20) clips shall be selected from each heat treated batch for test. If any clip fails to comply with the applicable requirement, the heat treated batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4. Samples complying with the requirement shall be returned to the lot.

4.3.3.5 Hardness of launcher positioning clip (see dwg. 9216145) - Major defect - Code No. 22001.- Twenty (20) launcher positioning clips shall be selected from each heat treated batch for test. If any clip fails to comply with the applicable requirement, the heat treated batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4. Samples complying with the requirement shall be returned to the lot.

4.3.3.6 Functioning (see 3.6) - Major defect - Code No. 23001.- Thirty (30) adapters shall be selected from each lot for this test. The test shall be performed as specified in 4.4.6. If more than one failure of the setback band to release the fuze lever occurs, or if more than three (3) adapters become detached from the grenade in flight, the lot shall be rejected. Samples used in this test shall not be returned to the lot.

4.3.3.7 Hydrostatic test of stabilizer tube (see dwg. 9216144) - Major defect - Code No. 24001.- This test shall be performed 100 percent (see 6.4). If any stabilizer tubing fails to comply with the applicable requirement, it shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.7, using equipment in accordance with 4.3.4. Samples complying with the requirement shall be returned to the lot.

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4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 9232012 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed in the ET shall be in accordance with MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.- The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 Strength of stabilizer tube brazed joint

4.4.1.1 1000 pound load test.- The brazed joint of the stabilizer tube assembly shall be tested by applying a load of 1000 pounds, min., to the inside of the brazed joint. The load shall be applied in a direction parallel to the axis of the tube at the rate of 0.100 to 0.125 inch per minute, inclusive. Observation shall be made for any evidence of joint failure. Test equipment shall be in accordance with 4.3.4.

4.4.1.2 7500 pound load test.- The stabilizer tube assembly shall be tested by applying a load in a direction parallel to the axis of the tube sufficient to cause failure in tension of the

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tube, or a load of 7500 pounds (whichever is reached first). The load shall be applied at a rate of 0.100 to 0.125 inch per minute, inclusive. The sides of the tube may be flattened, if necessary, for attachment to the testing machine. Observation shall be made to determine compliance with the applicable requirement. Test equipment shall be as specified in 4.3.4.

4.4.2 Strength of stabilizer tube and fin assembly welds.- The spot welded assemblies shall be pulled apart utilizing equipment in accordance with 4.3.4. Observation shall be made to determine compliance with the applicable requirement.

4.4.3 Embrittlement of springs.- The spring shall be repeatedly stretched with increasing loads until it does not return to its original free height. This point shall indicate minimum distortion. Any unit which fractures prior to extension to minimum distortion shall be classed defective.

4.4.4 Hardness of clip.- The clip shall be tested for hardness as specified in Federal Standard No. 151, Method 243.

4.4.5 Hardness of clip launcher position.- The clip launcher position shall be tested for hardness as specified in Federal Standard No. 151, Method 243.

4.4.6 Functioning.- The following tests shall be performed at a Government Proving Ground:

4.4.6.1 Setback band functioning test.- Select ten (10) adapters and ten (10) M14 incendiary hand grenades. Assemble each adapter to a grenade. Using a M7A3 Launcher, a M1 rifle, and M3 Grenade Cartridge and a sandbag butt rest, fire each assembly at 42 degrees to 45 degrees elevation with the stabilizer tube of each assembly covering only the outer four rings of the launcher.

4.4.6.2 Additional functioning tests.- Select twenty (20) adapters, ten (10) M8 (HC) Smoke Hand Grenades, and ten (10) M16 or M18 Colored Smoke Hand Grenades. Assemble ten (10) adapters to M8 Grenades and fire as in 4.4.6.1, except with the stabilizer tube in complete engagement with the launcher. Assemble the remaining ten (10) adapters to M16 or M18 Grenades and fire in the same manner. Observations in 4.4.6 shall be made to determine compliance with the applicable requirement.

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4.4.7 Hydrostatic test of stabilizer tube.- Prior to machining, the stabilizer tubing shall be tested for not less than five (5) seconds at the minimum hydrostatic pressure specified on the drawing. Observation shall be made for rupture, indication of flaws as evidenced by leaks, and for permanent deformation beyond the drawing requirements. Test equipment shall be as specified in 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.- The adapters shall be preserved and packaged in accordance with dwg. 9232107.

5.2 Packing

5.2.1 Level A.- The carton shall be packed in accordance with dwg. 9232107.

5.3 Marking.-

5.3.1 The carton and exterior containers shall be marked in accordance with dwgs. 9232107 and 9232108.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Submission of first article quantity.-Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out, may be obtained from the contracting officer.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Equipment Lists.-Equipment part or identifying lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 through 6.3.6. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.5 and 6.3.6 will be designated in the EL by the following codes. Other inspection equipment listed on the EL and not coded are defined in 6.3.4.

MU - Army design, mandatory for use.
CD - Contractor design and/or commercial equipment on contractor format. (Requires submission for Government approval as specified in 6.3.7).

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or

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commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Envelope drawing.-Envelope drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of envelope drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of envelope drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls:

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All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Hydrostatic test.- If the stabilizer tubing procured under the specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

6.5 Submission of test data and ammunition data cards.- In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-ND5
SMUPA-DC

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1330-A-109

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 OCT 64 1426

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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COMMANDING OFFICER
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Picatinny Arsenal
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